

Complete real networks: path dispersion and mapped storage reshape DOC signals

Three existing forms; 297 station-network instances / 295 outlets. Gold reaches mark mapped lakes/reservoirs; geometry-selected examples.

a Elongated / tributary-rich
n = 102 · 03030852



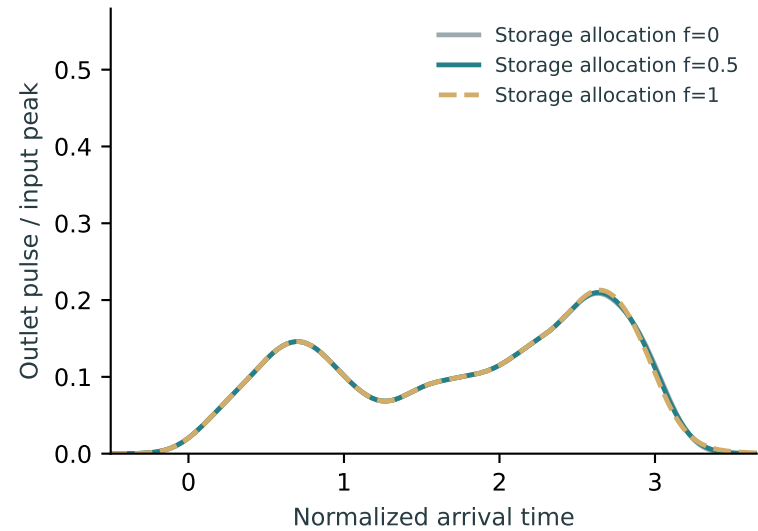
b Mainstem dominated / sparse
n = 30 · 06893990



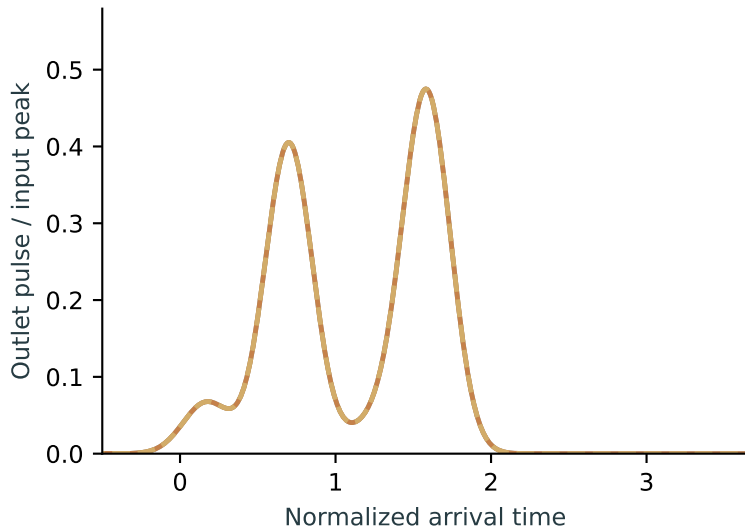
c Broad / tributary-rich
n = 165 · 07099200



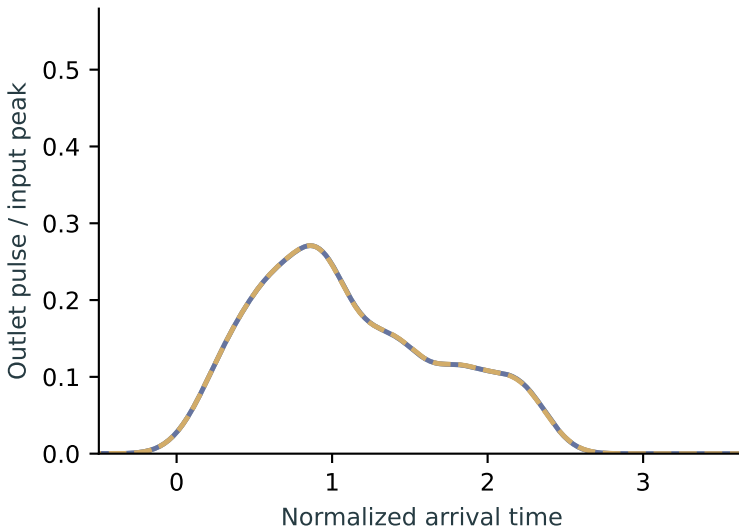
d Identical forcing, unchanged mean



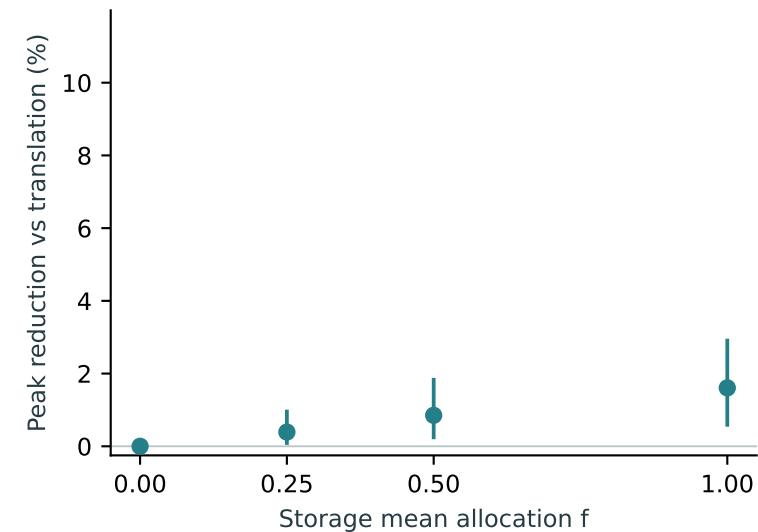
e Identical forcing, unchanged mean



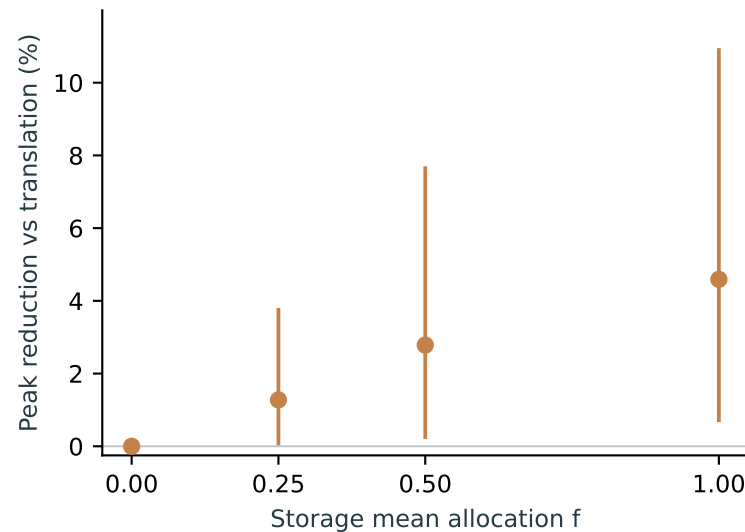
f Identical forcing, unchanged mean



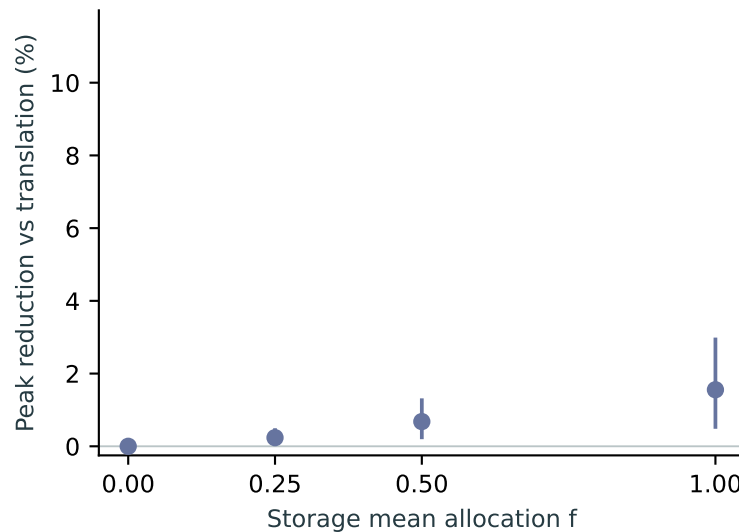
g All networks: mean and HUC4 interval



h All networks: mean and HUC4 interval



i All networks: mean and HUC4 interval



Identical DOC input in every incremental catchment; constant flow shares proportional to area. Time = channel length / $\sqrt{\text{basin area}}$, not measured days.
f redistributes translation into conservative storage at unchanged path mean. Bottom: HUC4-block bootstrap 95%; maps: independent scales, curves: shared axes.